WEEK 2 PROJECT REPORT

SAP SD Integration Architecture

Technical Blueprint for Sales, Inventory, Delivery and Finance

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Case Study: ABC Electronics Pvt. Ltd. (Hypothetical)

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Document Control

Field	Details
Document title	SAP SD Integration Architecture
Version	1.0 — improved submission draft
Business scenario	ABC Electronics Pvt. Ltd., a hypothetical electronics seller
Purpose	Design a proposed technical architecture integrating SAP SD with inventory, delivery, finance and external systems.
Status	Design proposal. Interface availability and configuration must be confirmed in the target landscape.

Executive Summary

This report proposes an integration architecture for the Order-to-Cash (O2C) process at ABC Electronics Pvt. Ltd. The design connects SAP Sales and Distribution (SD) with Inventory Management / Materials Management (MM), warehouse and shipping operations, and Financial Accounting (FI). It also considers an external customer or e-commerce channel and the use of middleware where external integration is needed.

The proposed architecture uses native SAP process integration for relevant SD, MM and FI activities within SAP S/4HANA, while APIs, IDocs and middleware may be used for external systems according to interface capabilities and business requirements. The design includes interface mapping, data flow, communication options, error handling, security controls, testing scenarios, measurable acceptance targets and an implementation roadmap.

This is a conceptual design for an internship assignment. It does not claim that the interfaces have been configured, deployed or tested in a live SAP environment. Targets in this report are proposed acceptance criteria and must be validated with business and technical stakeholders.

1. Introduction and Business Context

ABC Electronics Pvt. Ltd. sells consumer electronics through sales representatives and an online channel. The company needs a consistent flow from customer order capture through stock confirmation, delivery, goods issue, billing and payment clearing. Fragmented data or failed messages could cause delayed deliveries, incorrect stock visibility, billing errors or unreconciled financial postings.

The architecture focuses on a single logical Order-to-Cash process. Actual SAP configuration, organizational units, document types, availability rules, account determination, warehouse solution and external systems are not provided in the task; therefore, the report describes a proposed design and explicitly identifies assumptions.

1.1 Objectives

- Connect sales order processing with inventory, delivery and financial accounting processes.
- Define interface boundaries, data ownership, communication options and message triggers.
- Support timely transaction processing while using scheduled jobs for suitable non-urgent work.
- Provide controlled error handling, monitoring, security and data integrity measures.
- Define measurable test criteria to validate the design before any production deployment.

1.2 Scope and Assumptions

In scope	Out of scope / assumptions
SAP SD order, delivery and billing process; MM/inventory effects; FI postings; optional external customer and warehouse interfaces.	No live SAP configuration, middleware deployment, credentials or production connectivity is included.
Proposed APIs, IDoc or middleware patterns where appropriate; monitoring, security and testing design.	Specific API names, IDoc message types, endpoints and fields must be confirmed against the target SAP release and system design.
Hypothetical company and illustrative performance targets.	No test execution results or measured business outcomes are claimed.

2. Business Requirements and Current-State Analysis

For this design exercise, the assumed current state is that order information may be entered through more than one channel, stock status may be checked separately, and shipment and billing status may not be visible in one end-to-end view. These are scenario assumptions for the case study, not verified facts about a real company.

ID	Requirement / challenge	Proposed response	Evidence to collect during discovery
BR-01	Reduce repeated manual entry of customer orders.	Validate and transmit order data through a controlled interface.	Sample order forms; number of manual re-entries.
BR-02	Improve stock and delivery visibility.	Use SAP availability and delivery status; exchange warehouse updates if external WMS exists.	Stock check examples; delivery status and timestamps.
BR-03	Ensure billing creates expected financial postings.	Use standard SAP billing-to-accounting integration and monitor posting status.	Billing document and linked accounting document in test system.
BR-04	Identify and resolve interface failures quickly.	Log messages, classify errors, alert support and safely retry transient failures.	Error log fields; support ownership and response target.
BR-05	Avoid duplicate or incomplete transactions.	Use unique external message/order references, validation and reconciliation.	Duplicate-message test; count of unmatched order/delivery/billing records.

2.1 Proposed Success Measures

The following are initial design targets for discussion and acceptance testing. They are not measured results and should be agreed with the project owner before implementation.

Metric	Proposed target	How to measure
Valid interface processing success	At least 99% during an agreed test window	Successful valid messages ÷ total valid messages received × 100.
Synchronous external order response	95% of test requests within 5 seconds, under agreed test load	Capture request and response timestamps during load testing.
Duplicate-order control	Zero unintended duplicate orders in the defined duplicate-message test set	Replay each test message and verify order count and reference handling.
Critical data validation	100% of defined invalid mandatory-field cases rejected or routed to a controlled error	Execute negative test cases and compare actual results to expected outcomes.
Reconciliation	All test orders traceable to applicable delivery, billing and accounting status	Compare document references and status across the test sample.

3. Proposed Integration Architecture

The architecture separates the SAP core process from external channels. SAP S/4HANA is the system of record for the configured sales, inventory and accounting documents. External systems may connect through a managed integration layer when required. Middleware is not assumed to be necessary for every internal SD–MM–FI interaction.

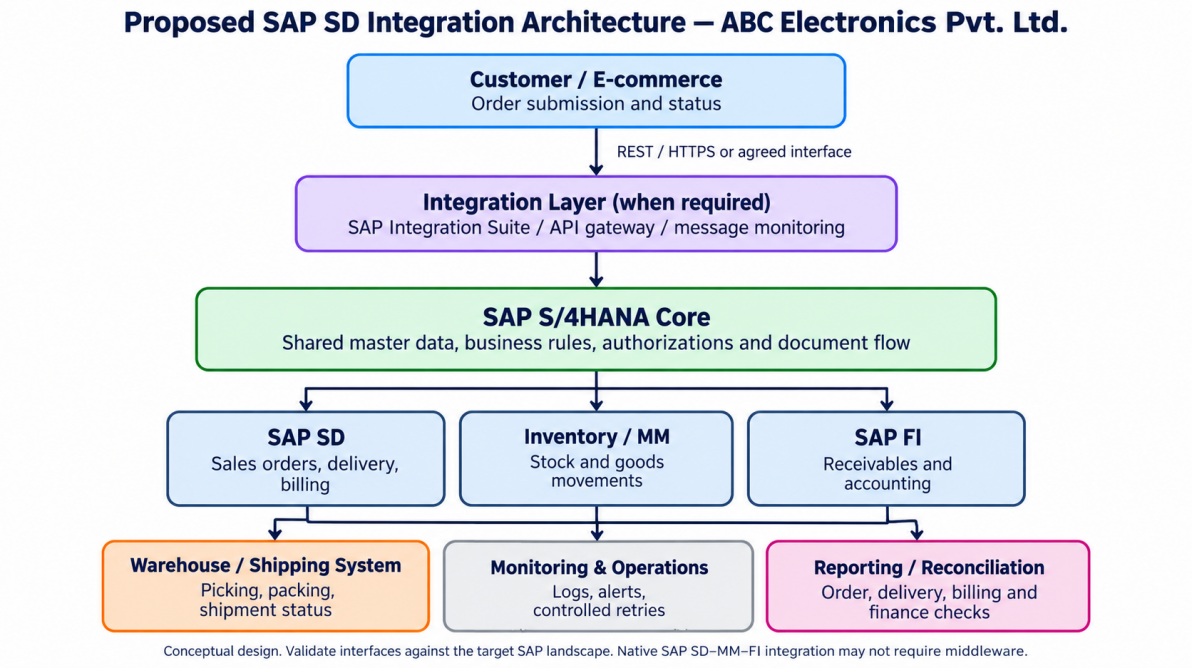


Figure 1. *Proposed architecture. The integration layer is conditional and is intended mainly for external systems, routing, mapping, monitoring or protocol mediation.*

3.1 Component Responsibilities

Component	Responsibility	Key data / outputs
SAP SD	Sales order processing, delivery processing and billing.	Sales orders, delivery documents, billing documents and statuses.
SAP MM / Inventory Management	Material and stock management; goods movements.	Material/plant/stock data, goods issue and movement status.
SAP FI	Financial accounting entries and customer open items.	Receivables, revenue/tax postings and accounting document references.
Warehouse / Shipping System (optional external)	Picking, packing, dispatch and shipment events.	Warehouse tasks, shipment references, delivery confirmations.
Customer / E-commerce Channel (optional)	Capture customer orders and receive acknowledgements/status.	External order ID, customer, items, quantity, requested date.
Integration Layer (conditional)	Routing, mapping, protocol mediation, monitoring and controlled retries.	Message IDs, processing statuses, error records.
Operations / Reconciliation	Monitor exceptions and compare business document status.	Alerts, reconciliation results, incident and audit records.

4. Integration Interface Mapping

The interface choices below are proposed patterns. Final interface contracts must be selected after confirming SAP S/4HANA release, deployment model, available APIs, middleware, warehouse system and non-functional requirements.

ID	Source → Destination	Data exchanged	Proposed interface / trigger	Error and control
IF-01	Customer channel → SAP SD	External order ID, sold-to party, ship-to party, material, quantity, requested date, purchase-order reference.	REST API over HTTPS or an approved SAP interface; triggered when an order is submitted.	Validate mandatory fields; return accepted/rejected status; record external ID to detect duplicates.
IF-02	SAP SD ↔ SAP MM / Inventory	Material, plant, requested quantity, availability and goods movement references.	Native SAP process integration; availability check and goods movement at configured process steps.	Use standard SAP checks and document status; monitor failed postings or incomplete documents.
IF-03	SAP SD → Warehouse / Shipping	Delivery number, items, quantities, ship-to details, picking/packing instructions.	API or IDoc-based integration, if an external warehouse system is in scope; trigger on delivery creation/release.	Track message ID and acknowledgement; retry transient failures safely; reconcile delivery status.
IF-04	Warehouse → SAP	Picking status, packed quantity, shipment/tracking reference, dispatch status.	API, IDoc or approved event interface, depending on warehouse capability.	Validate delivery reference and quantity; reject stale or duplicate updates.
IF-05	SAP SD Billing → SAP FI	Billing amount, customer receivable, revenue/tax account determination references.	Native SAP billing-to-accounting integration; triggered during billing posting.	Monitor accounting status; resolve account-determination or posting errors before reprocessing.
IF-06	SAP / external reporting → Reconciliation	Order, delivery, billing, accounting status and reference IDs.	Scheduled report, approved API or batch export, depending on reporting design.	Compare counts and totals; record missing, duplicated or unmatched references.

4.1 Interface Contract Fields

For each interface, the technical design should specify: interface owner; source and destination; business purpose; message structure; mandatory and optional fields; field mapping; data types and lengths; authentication; transport; trigger; expected response; timeout; retry policy; idempotency key; logging fields; monitoring owner; and retention period.

Example incoming order mapping: externalOrderId → external reference; customerCode → sold-to party; shipToCode → ship-to party; item.materialCode → SAP material; item.quantity → order quantity; requestedDeliveryDate → requested date. Exact SAP field names and API payload structures must be confirmed from the selected interface documentation rather than assumed.

5. Order-to-Cash Data Flow



Figure 2. *Proposed transaction flow. The sequence is illustrative and depends on configuration, availability, delivery eligibility, billing prerequisites and accounting settings.*

5.1 Narrative of the Transaction

1. **Order capture:** the customer submits an order through a sales representative or external channel. The interface validates required fields and preserves the external order reference.
2. **Sales order:** SAP SD validates relevant customer, material, sales-area, pricing and order data. A sales order is created only if the applicable checks pass.
3. **Availability and delivery:** SAP checks availability according to configuration. When delivery is due and eligible, an outbound delivery is created and warehouse execution can begin.
4. **Goods issue:** after the required picking and shipping steps, posting goods issue updates inventory and may generate relevant accounting effects according to configuration.
5. **Billing and FI:** billing is created when prerequisites are satisfied. Billing normally transfers relevant receivable, revenue and tax postings to Financial Accounting when the accounting posting succeeds.
6. **Payment and clearing:** incoming payment is recorded through the configured finance process and the customer open item is cleared.
7. **Monitoring and reconciliation:** the support team traces order, delivery, goods issue, billing and accounting references to identify incomplete or inconsistent processing.

Illustrative SAP GUI transaction codes often used in training include VA01 (create sales order), VA03 (display sales order), VL01N (create outbound delivery) and VF01 (create billing document). Availability and permissions depend on the training system. A transaction code is not an integration interface by itself.

6. Communication Protocols and Technical Specifications

Technology	Typical use in this design	Design considerations
HTTPS / TLS	Secure transport for external APIs and web services.	Use current approved TLS settings, certificate validation and secure endpoint configuration.
REST API / JSON	Synchronous order submission or status lookup where supported.	Define schema, mandatory fields, response codes, timeout, versioning and idempotency.
IDoc / ALE	Structured asynchronous SAP message exchange where supported and appropriate.	Confirm message type, basic type/extension, partner profile, processing status and reprocessing procedure.
OData / SAP APIs	Standardized access to SAP business data/processes where a suitable released API exists.	Select the API based on the exact SAP product/release and permitted use; do not assume every object is exposed.
Scheduled batch / file	Periodic reconciliation or non-urgent bulk exchange.	Define file format, naming, encryption, control totals, schedule, retention and restart behaviour.
Middleware mapping and routing	Transform and route external messages where systems use different schemas or protocols.	Version mappings, monitor messages, protect credentials and prevent uncontrolled retries.

6.1 Proposed Non-Functional Specifications

Area	Proposed requirement
Availability	Agree service hours and availability target with business owner; define planned maintenance handling.
Performance	Validate the illustrative 5-second 95th-percentile synchronous response target under agreed load.
Scalability	Test expected average and peak message volumes; avoid hard-coded assumptions until volumes are known.
Observability	Capture correlation ID, external reference, timestamp, processing status, error category and retry count.
Recoverability	Document retry, replay, reconciliation and manual recovery procedures; protect against duplicate processing.
Change management	Version interface schemas and mappings; test backward compatibility and obtain approvals before deployment.

7. Middleware and Processing Strategy

SAP Integration Suite is one possible middleware option in an SAP-centric landscape. A different approved integration platform may be selected based on licensing, existing enterprise standards and connected systems. Middleware is proposed when external systems need protocol mediation, message transformation, routing, monitoring or decoupled processing.

7.1 Real-Time vs Batch

Processing style	Suitable example	Benefits	Risks / controls
Synchronous / near real-time	Submit an order and return an immediate acceptance or validation response.	Fast feedback for the customer channel.	Timeouts can create uncertainty; use correlation IDs and idempotency to avoid duplicate orders.
Asynchronous message processing	Send delivery updates or larger messages where the sender need not wait for final processing.	Decouples sender and receiver; supports queues and controlled retries.	Requires message monitoring, acknowledgement and dead-letter/manual review handling.
Scheduled batch	Daily reconciliation, bulk updates or non-urgent reports.	Efficient for high-volume non-urgent processing.	Define schedule, control totals, restart point, duplicate handling and missed-run alerts.

The design should choose processing style by business need rather than force every integration into real-time processing. Stock and order responses may require prompt feedback, while reconciliation reports can normally be scheduled.

8. Error Handling and Monitoring

Error scenario	Detection / response	Recovery and ownership
Missing or invalid customer/material	Reject or stop processing with a clear validation message; record field and reason.	Sales support corrects master data or order input; resend only after validation.
Insufficient stock	Return configured availability result or create a shortage/backorder path if enabled.	Sales/warehouse team follows business policy; do not create an unsupported stock movement.
Network timeout / transient external failure	Log message and correlation ID; alert or mark pending.	Retry with bounded backoff where safe; check whether the original request succeeded before replay.
Duplicate message	Compare external order ID/idempotency key and processing status.	Return existing result or flag duplicate; do not create a second order unintentionally.
Warehouse update references unknown delivery	Reject/quarantine update and record delivery reference.	Operations verifies source and delivery; correct mapping/reference and reprocess under control.
Billing accounting posting failure	Monitor billing/accounting status and relevant application log.	Finance/support corrects configuration or master data and reprocesses through approved procedure.
Batch job failure	Alert on job status, missing output or control-total mismatch.	Operations restarts from a documented checkpoint and reconciles totals.

8.1 Monitoring Fields

- Correlation/message ID and external order reference.
- Source, destination, interface name and schema version.
- Created/received timestamp, processing status and duration.
- Error category, technical message, retry count and next action.
- Business document references: sales order, delivery, material document, billing and accounting document where applicable.

Logs should avoid storing unnecessary personal or sensitive data. Access to detailed payloads should be restricted and retention should follow the organisation's approved policy.

9. Security and Data Integrity

Control	Implementation approach
Identity and access	Use role-based access and least privilege for users, technical users and interface accounts; review access periodically.
Transport protection	Use HTTPS/TLS for external connections and validate certificates; use approved secure transfer methods for files.
Authentication and secrets	Use approved authentication such as OAuth or certificates where supported; store secrets in managed secure storage, not source code or plain-text documents.
Input validation	Validate mandatory fields, customer/material references, quantities, dates and permitted values before creating business documents.
Duplicate prevention	Use a unique external message/order reference and idempotent processing where feasible; define duplicate response behaviour.
Auditability	Keep controlled logs of interface actions, status changes and reprocessing decisions; restrict log access.
Data minimization and privacy	Exchange only necessary fields; apply retention, masking and applicable privacy requirements.
Segregation of duties	Separate configuration, approval and production support responsibilities where feasible; require change approval.
Reconciliation	Compare message counts and business document references across order, delivery, billing and accounting stages.

Compliance requirements depend on the company's jurisdictions, industry and internal policies. The project team should confirm applicable privacy, tax, financial-record retention and security obligations with the responsible stakeholders; this report does not assert a specific certification or legal compliance status.

10. Testing Plan and Acceptance Criteria

The following test plan is proposed for a future test environment. Test results are not claimed as executed. The test owner should record actual result, evidence link, defect ID and sign-off for each case.

Test ID	Precondition / scenario	Test action	Expected result / acceptance criterion
T01	Valid customer/material and sufficient stock.	Submit a valid order through the selected channel or create it in SAP GUI.	Order accepted once; key fields and external reference match input.
T02	Customer ID is invalid or missing.	Submit order with invalid customer reference.	Order rejected or held with clear error; no unintended valid sales order created.
T03	Material ID invalid or mandatory item data missing.	Submit malformed item data.	Validation error identifies the issue; no incomplete order is processed.
T04	Stock is insufficient.	Enter order quantity above available quantity.	Configured availability result is returned; no unsupported goods issue occurs.
T05	Delivery is eligible and warehouse steps completed.	Create delivery, pick/pack as configured and post goods issue in test system.	Delivery and goods movement statuses update; stock and relevant postings are consistent.
T06	Billing prerequisites satisfied.	Create billing document for the eligible delivery/order.	Billing document created; expected accounting posting status and document reference available.

Test ID	Precondition / scenario	Test action	Expected result / acceptance criterion
T07	External request times out after submission.	Simulate timeout and resend same external order reference.	No unintended duplicate order; original outcome can be identified or reconciled.
T08	Temporary interface failure.	Simulate transient endpoint/message failure.	Failure is logged; bounded retry or controlled reprocessing works and is auditable.
T09	Warehouse update has an unknown delivery number.	Send a status update with invalid reference.	Message is rejected/quarantined with reason; valid records remain unchanged.
T10	Unauthorized user or interface identity.	Attempt access without required permission.	Request denied and security event logged.
T11	Batch reconciliation runs on sample records.	Run scheduled comparison on known order/delivery/billing references.	All expected records reconcile; deliberately missing/duplicate record is flagged.
T12	Peak-volume test set is agreed.	Run load test and measure latency/success rate.	Proposed target met: ≥99% valid message success and 95th-percentile synchronous response ≤5 seconds, or deviations formally approved.

10.1 Test Evidence Template

Test ID	Actual result	Pass / Fail / Not run	Evidence / defect reference	Tester / date
T01	To be completed during execution	Not run	Not applicable yet	Assigned tester
T02	To be completed during execution	Not run	Not applicable yet	Assigned tester
T03	To be completed during execution	Not run	Not applicable yet	Assigned tester
T04–T12	Record each test separately when executed	Not run	Not applicable yet	Assigned tester

Do not mark a test as passed until it has been executed and evidence has been reviewed. If the internship task is design-only, the table demonstrates the proposed validation plan.

11. Implementation Roadmap

Phase	Indicative activities	Deliverables / gate
1. Discovery	Confirm systems, business process, order volumes, master data ownership, security and support owners.	Approved requirements and assumptions register.
2. Detailed design	Confirm interface contracts, field mappings, protocols, error codes, security and non-functional targets.	Architecture diagram, interface specification and design review approval.
3. Build / configuration	Configure selected APIs, IDocs, middleware flows, mappings and monitoring in a non-production landscape.	Configured interfaces and deployment notes.
4. System and integration testing	Execute positive, negative, retry, duplicate, security and performance tests.	Test evidence, defect log and remediation plan.
5. User acceptance testing	Business users validate order-to-cash scenarios and reconcile documents.	Business sign-off and accepted known limitations.
6. Deployment readiness	Review access, monitoring, rollback, support contacts, cutover and recovery plans.	Go-live checklist and rollback plan.
7. Hypercare and improvement	Monitor transaction health, review errors and tune performance.	Operational handover, KPI report and improvement backlog.

Durations should be estimated only after scope, team availability and system access are known. A project schedule should include dependencies, accountable owners and formal approval gates.

12. Risks and Mitigations

Risk	Impact	Mitigation
Unconfirmed API or interface availability	Design may not be implementable as drawn.	Validate exact SAP release, deployment model and released interfaces during discovery.
Poor master-data quality	Orders or deliveries fail validation.	Define data owners, validation rules and master-data cleanup before testing.
Duplicate messages during retries	Duplicate sales orders or status changes.	Use unique references/idempotency, message tracking and reconciliation.
Unclear ownership of failed messages	Slow recovery and unresolved business exceptions.	Define a support RACI, alert routing, response targets and escalation path.
Security misconfiguration	Unauthorized access or data exposure.	Apply least privilege, secure credentials, transport encryption and access reviews.
Performance targets not realistic	Slow response during peak traffic.	Agree volumes and service targets; test under representative load and revise based on evidence.
Incorrect accounting determination	Billing cannot post or posts incorrectly.	Validate configuration and account determination with SAP FI stakeholders in a test system.

13. Roles and Responsibilities

Role	Responsibility
Business / Sales owner	Confirm sales requirements, order rules, customer impact and acceptance criteria.
SAP SD functional consultant	Define sales order, delivery and billing process requirements; validate SD document flow.
SAP MM / Inventory consultant	Validate stock checks, goods movements and material/plant dependencies.
SAP FI consultant	Validate billing accounting integration, account determination and reconciliation.
Integration developer / architect	Define API/IDoc/middleware contracts, mapping, routing, security and technical monitoring.
Security / Basis / Platform team	Approve technical identities, connectivity, certificates, authorizations and operational monitoring.
Test lead / business tester	Execute test cases, record evidence, manage defects and obtain sign-off.
Operations / support	Monitor messages and jobs, resolve incidents and carry out controlled recovery.

14. Conclusion

This proposed architecture describes how SAP SD can participate in an end-to-end Order-to-Cash process integrated with inventory, warehouse operations and Financial Accounting. It separates native SAP process integration from optional external interfaces, selects communication patterns based on business need, and defines error handling, security, monitoring and test criteria.

The next step before implementation is to validate the assumed landscape, select supported interfaces for the target SAP release, agree non-functional targets with stakeholders, and execute the documented test plan in an authorized non-production environment. The architecture is a design proposal and should not be interpreted as evidence of a deployed or tested integration.

15. References

- **SAP Help Portal —** https://help.sap.com/ — Use the exact SAP S/4HANA product version and search for sales order processing, outbound delivery, billing, goods issue, accounting integration and released APIs.
- **SAP Learning —** https://learning.sap.com/ — Use learning journeys for SAP S/4HANA Sales and related business processes.
- **SAP Integration Suite documentation —** https://help.sap.com/docs/integration-suite — Consult the applicable documentation for integration flows, monitoring, security and supported adapters if Integration Suite is selected.
- **SAP API Business Hub / SAP Business Accelerator Hub —** https://api.sap.com/ — Verify released API specifications and payloads for the target SAP product and release.

Reference note: *consult and cite the exact pages used in the final submission. This document provides reference portals, not claims that a particular API or adapter has already been verified for the hypothetical landscape.*

Appendix A: Glossary

Term	Meaning
ACDOCA	Universal Journal table in S/4HANA holding all financial line items
aATP	Advanced Available-to-Promise: availability check in S/4HANA
ALE / IDoc	Application Link Enabling / Intermediate Document: SAP asynchronous message format
BAPI	Business Application Programming Interface (stable function module interface)
BP / CVI	Business Partner / Customer-Vendor Integration
COGS	Cost of Goods Sold
COPA	Profitability Analysis in Controlling
EWM	Extended Warehouse Management
FSCM	Financial Supply Chain Management (includes Credit Management)
GI	Goods Issue
iFlow	Integration flow in SAP Integration Suite
MATDOC	Material document table in S/4HANA
OBYC / VKOA	Transactions for inventory and revenue account determination
OTC	Order-to-Cash
RAR	Revenue Accounting and Reporting
SoD	Segregation of Duties
VBFA	Sales document flow table